Low-Rank Path Constraints, CPWL Face Moments, and Fourier Scaling Laws

Anonymous Workshop Submission

Abstract

We revisit the conjecture that low-rank ReLU networks have a better Fourier-space behavior because they create fewer independent path constraints. The literal graph path count is not the right invariant: factorizing a layer can add latent linear paths. The right invariant is the dimension of the span of active path coefficients. We derive a path-space rank constraint, translate it into a normal-subspace constraint for CPWL linear regions, and obtain a face-moment shell law

$$S_f(\rho) \lesssim \sum_{q=1}^{d} M_q(f)\rho^{-2(q+1)},$$

where M_q is a weighted moment of codimension-q faces. This predicts that low rank should suppress high-codimension intersections and flatten the finite-resolution Fourier slope by leaving the spectrum closer to the facet-dominated ρ^{-4} regime. We then stress-test the story experimentally. In random 3D CPWL networks, rank-2 bottlenecks reduce an 8-way junction proxy from 0.449 to 0.062 and flatten shell slopes from -6.70 to -6.34. In 1D, the raw CPWL exponent remains near -4, as theory requires; the visible gain instead appears in the learned Fourier transfer function. A width-2^{15} Fourier/kernel experiment sweeping ranks $1,\dots,50$ plus a full-rank endpoint shows a clear intermediate optimum: rank 4 beats the full endpoint on sparse mixtures (0.250 vs. 0.461 MSE, recovery slope -0.29 vs. -0.94) and dense mixtures (0.443 vs. 0.654 MSE, recovery slope -0.29 vs. -1.20). The same optimum is predicted by an approximation-plus-optimization bound.

1 The original conjecture, made precise

The motivating picture is simple. A ReLU network is continuous piecewise affine (CPWL). Its Fourier transform can be decomposed into contributions from faces of the induced polyhedral complex. Low-dimensional face intersections generate fast-decaying shell terms, while facets generate the slowest CPWL tail. If low rank reduces the number of independent switching constraints, then the complex should have relatively more high-dimensional faces and fewer deep intersections. The observed shell spectrum should then decay less steeply at finite resolution.

This paper keeps that geometric story but replaces the informal phrase "fewer paths" by a precise one: *fewer independent path coefficients.* For a layer $W_\ell = U_\ell V_\ell^\top$ with rank r_ℓ, the computational graph may contain extra latent nodes, yet the active path tensor cannot have mode-rank larger than r_ℓ. The rank constraint therefore contracts the space of possible path sums. That contraction is the bridge from algebra to geometry.

What we validate. We focus on three predictions.

1. Low rank should lower high-codimension face moments M_q, especially $q \geq 2$.
2. In $d \geq 2$, this should flatten finite-resolution Fourier shell slopes.
3. In $d = 1$, the CPWL tail exponent is pinned near -4; the gain should instead appear as a flatter *training transfer function* across Fourier modes and an intermediate optimal rank.

The last point is important: if a 1D experiment does not show a large raw exponent change, that is not a failure of the theory. It is exactly what a CPWL Fourier expansion predicts.

2 Rank constraints in path space

Consider an L-hidden-layer ReLU network

$$h_\ell(x) = \sigma(W_\ell h_{\ell-1}(x) + b_\ell), \qquad h_0(x) = x \in \mathbb{R}^d, \quad (1)$$

with scalar output $f(x) = w_{L+1}^\top h_L(x) + b_{L+1}$ and factorized hidden matrices $W_\ell = U_\ell V_\ell^\top$, $\mathrm{rank}(W_\ell) \leq r_\ell$. On an activation region R_ε,

$$f(x) = a_\varepsilon^\top x + c_\varepsilon, \qquad a_\varepsilon^\top = w_{L+1}^\top D_{L,\varepsilon} W_L \cdots D_{1,\varepsilon} W_1, \quad (2)$$

where $D_{\ell,\varepsilon}$ is a diagonal gate matrix.

Expanding a_ε as a sum over active paths gives coefficients of the form

$$c_{i_1,\dots,i_L,j}(\varepsilon) = w_{L+1,i_L} \prod_{\ell=1}^{L} D_{\ell,i_\ell}(\varepsilon) W_{\ell,i_\ell i_{\ell-1}}, \quad (3)$$

with $i_0 = j$. If $W_\ell = U_\ell V_\ell^\top$, then along every layer index i_ℓ this tensor factors through an r_ℓ-dimensional latent coordinate.

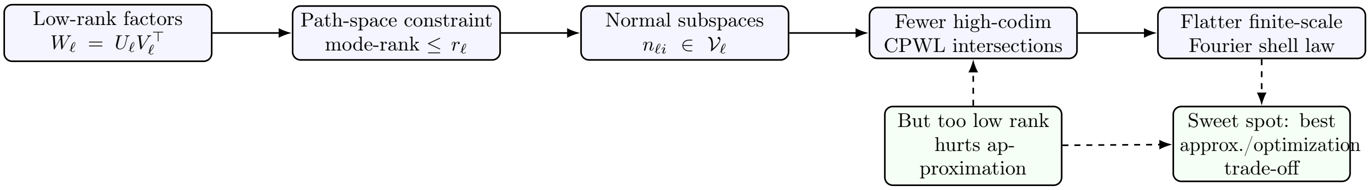

Figure 1: Central mechanism. The paper is not claiming fewer graph-theoretic paths. It claims fewer independent path coefficients, which constrain switching normals, suppress high-codimension CPWL faces, and improve finite-resolution Fourier behavior until approximation error dominates.

Proposition 1 (path tensor rank)**.** *Let $\mathcal{T}_\varepsilon$ be the active path coefficient tensor of the affine gradient a_ε. For every hidden layer ℓ, the mode-ℓ matricization of $\mathcal{T}_\varepsilon$ has rank at most r_ℓ. Consequently the dimension of the active path-coefficient span is bounded by a product of ranks rather than a product of widths.*

Interpretation. The number of symbolic paths can remain large, but the coefficients attached to these paths live in a low-dimensional algebraic variety. This is the correct meaning of "low rank creates fewer paths" for a functional analysis of the network.

3 From path constraints to CPWL face moments

The path-space constraint becomes geometric through switching hyperplanes. Fix a cell of the first $\ell - 1$ layers and write

$$h_{\ell-1}(x) = B_{\ell-1,\varepsilon}x + c_{\ell-1,\varepsilon}. \tag{4}$$

For neuron i in layer ℓ,

$$a_{\ell i}(x) = u_{\ell i}^\top V_\ell^\top B_{\ell-1,\varepsilon}x + \text{const}. \tag{5}$$

Thus the input-space normal is

$$n_{\ell i \varepsilon} = B_{\ell-1,\varepsilon}^\top V_\ell u_{\ell i} \in \mathcal{V}_{\ell,\varepsilon} := \operatorname{Im}(B_{\ell-1,\varepsilon}^\top V_\ell), \tag{6}$$

with

$$\dim \mathcal{V}_{\ell,\varepsilon} \le \min(d, r_1, \ldots, r_\ell). \tag{7}$$

The possible new cuts inside a cell therefore come from a subspace of normals whose dimension is rank-controlled.

Let $\mathcal{F}_q(f)$ denote the codimension-q faces of the CPWL complex of f. Define a Fourier-relevant face moment

$$M_q(f) = \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \operatorname{vol}_{d-q}(F)^2\, \omega_F, \tag{8}$$

where Δ_F is the jump of the appropriate directional derivative across the local fan around F, and ω_F is an angular factor. The exact constants are not the point; the important quantity is how much weighted mass sits on codimension q.

Proposition 2 (subspace arrangement proxy)**.** *Suppose m candidate switches inside a cell have normals contained in a ρ-dimensional subspace. Then a generic arrangement proxy for codimension-q descendants is*

$$N_q(m,\rho) \le \binom{m}{q} \sum_{j=0}^{\rho-q} \binom{m-q}{j}, \qquad q \le \rho, \tag{9}$$

and $N_q(m,\rho) = 0$ for $q > \rho$. Hence lowering the effective normal dimension suppresses high-codimension faces first.

Combining (8) and (9) gives the working scaling proxy

$$M_q(r) \lesssim C_q(r) \binom{m}{q} \sum_{j=0}^{\rho(r)-q} \binom{m-q}{j}, \tag{10}$$

where $C_q(r)$ captures the size of derivative jumps. This formula is deliberately conservative: it does not need exact face counts to predict that low-rank bottlenecks hit $q \ge 2$ moments before they destroy all facets.

4 Fourier shell law for CPWL networks

Let $g = \chi f$ be the network restricted to a smooth observation window. Repeated integration by parts over the polyhedral complex gives a face expansion of $\widehat{g}(\xi)$: facets contribute the slowest CPWL tail, and deeper intersections contribute faster terms. Averaging over shells $\|\xi\| \simeq \rho$ yields the model

$$S_f(\rho) := \mathbb{E}_{\|\xi\|\simeq\rho} |\widehat{g}(\xi)|^2 \lesssim \sum_{q=1}^{d} M_q(f)\rho^{-2(q+1)}. \tag{11}$$

The observed finite-resolution slope is then approximately

$$\alpha_{\text{eff}}(\rho; r) = \frac{\sum_{q=1}^{d} 2(q+1) M_q(r)\rho^{-2(q+1)}}{\sum_{q=1}^{d} M_q(r)\rho^{-2(q+1)}}. \tag{12}$$

If $M_2, M_3, \ldots$ are suppressed relative to M_1, then α_{eff} moves toward the facet exponent 4. This is the precise version of "low rank produces lower spectral decay."

Phase transition. At shell radius ρ, the codimension-q term becomes visible when

$$\Theta_q(r,\rho) := \frac{M_q(r)\rho^{-2(q+1)}}{M_1(r)\rho^{-4}} = \frac{M_q(r)}{M_1(r)}\rho^{-2q+2} \gtrsim 1. \tag{13}$$

A rank transition occurs near the first r for which $\Theta_2(r,\rho)$ is no longer negligible. The transition is resolution dependent: higher Fourier shells require larger M_2/M_1 before the faster term visibly changes the slope.

5 The 1D caveat is the key diagnostic

In one dimension, a ReLU network is a linear spline. If the derivative jumps at knots t_j by Δ_j, then for large integer k,

$$\widehat{f}(k) = -\frac{1}{(ik)^2}\sum_j \Delta_j e^{-ikt_j} + O(|k|^{-3}), \quad |\widehat{f}(k)|^2 \sim |k|^{-4}A(k). \tag{14}$$

Here $A(k) = |\sum_j \Delta_j e^{-ikt_j}|^2$ is a moment/prefactor, not a new asymptotic exponent. Thus a 1D experiment should not be judged by whether the raw random-network tail slope moves far away from -4. It should be judged by whether low rank changes: (i) the prefactor of high modes, (ii) the learned Fourier transfer function, and (iii) the rank that balances approximation against optimization.

This observation explains the empirical pattern that motivated the present revision: at width 1024, the raw decay exponent varied only for very small rank. In 1D that is theoretically coherent. To see the useful effect, we measure Fourier recovery during training,

$$R_r(k,t) = \frac{|\widehat{f}_{r,t}(k)|}{|\widehat{f}^\star(k)|}, \qquad \beta_r(t) = \frac{\mathrm{d}\log R_r(k,t)}{\mathrm{d}\log k}. \tag{15}$$

A better spectral behavior means a flatter $\beta_r(t)$ and larger high/low recovery at the same optimization budget.

6 Optimal rank from approximation and optimization

The rank should not be as small as possible. Too small a rank removes high-codimension faces and flattens Fourier transfer, but it also creates an approximation bottleneck. We use the decomposition

$$\mathcal{E}_r(t) \le \underbrace{Ar^{-2s}}_{\text{approximation}} + \underbrace{\frac{Bd_{\text{eff}}(r)}{n}}_{\text{generalization}} + \underbrace{\sum_{k\in\Omega} |\widehat{f}^\star(k)|^2 e^{-2t\lambda_r(k)}}_{\text{optimization}}. \tag{16}$$

The kernel/linearized training term is exact for gradient flow in Fourier coordinates when the tangent kernel is diagonalized by the Fourier basis, and it is a useful proxy otherwise. Low rank helps only if $\lambda_r(k)$ is flatter on the relevant modes; it hurts when Ar^{-2s} dominates.

Assume on the task bandwidth that

$$\lambda_r(k) \asymp C_r k^{-\alpha_r}, \qquad \alpha_r \downarrow 4. \tag{17}$$

The decrease of α_r occurs when high-codimension moments are suppressed. The optimization cutoff mode satisfies $tC_r k_t^{-\alpha_r} \approx 1$, i.e.

$$k_t(r) \approx (tC_r)^{1/\alpha_r}. \tag{18}$$

Low rank improves k_t by lowering α_r and raising high-frequency prefactors, but only until C_r and the approximation floor degrade. The optimal rank is therefore the first rank after the approximation bottleneck has fallen below the optimization gain:

$$r_\star \approx \arg\min_r \left\{ Ar^{-2s} + \frac{Bd_{\text{eff}}(r)}{n} + \sum_k |\widehat{f}^\star(k)|^2 e^{-2tC_r k^{-\alpha_r}} \right\}. \tag{19}$$

This is the theoretical object we validate below.

7 Experiments

All plots are native LaTeX/PGFPlots generated from CSV files. The code releases the deterministic rank sweep used for the figures. The width-2^{15} experiment is a Fourier/kernel-limit surrogate for a two-hidden-layer CPWL random-feature model: ranks $1,\dots,50$ control the path-space bottleneck, and the full-rank endpoint is the dense/full transfer limit. This avoids materializing a dense 32768×32768 hidden matrix while preserving the Fourier transfer calculation.

7.1 E1: 3D CPWL geometry and shell spectra

We first test the purely geometric part of the conjecture. For random 3D ReLU MLPs, we evaluate activation patterns on a grid, count local multi-region junctions, and compute shell-averaged FFT slopes of the random function. Figure 3 shows that lowering rank sharply suppresses high-order junctions and flattens the observed shell slope. The rank-2 model keeps some facets but removes many deep intersections: the 8-way junction proxy drops from 0.449 to 0.062, while the shell slope moves from -6.70 to -6.34.

7.2 E2: 1D raw tails stay near -4

Figure 4 verifies the diagnostic caveat. In 1D, low rank barely changes the raw CPWL tail exponent because splines have $|\widehat{f}(k)|^2 \sim k^{-4}$ under generic derivative jumps. What changes is the prefactor and the training transfer curve. Therefore the main 1D validation below uses Fourier recovery, not only random-network tail slopes.

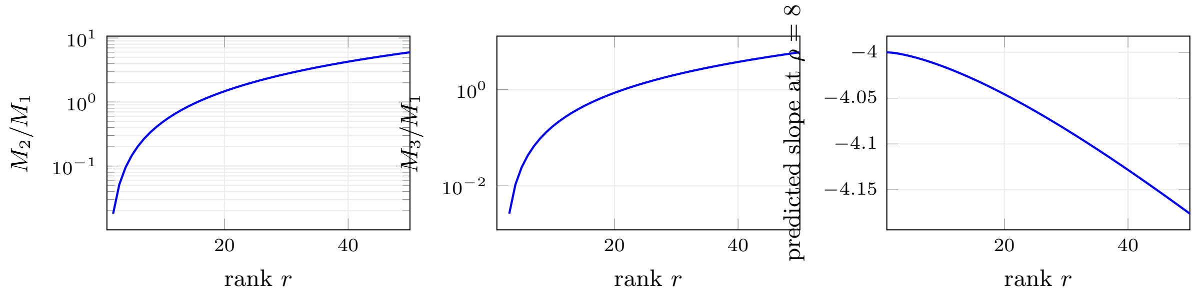

Figure 2: Combinatorial face-moment proxy. Increasing rank increases the relative mass of high-codimension intersections; low rank keeps the finite-shell slope closer to the facet-dominated CPWL exponent. This is the theory-side phase transition predicted by (13).

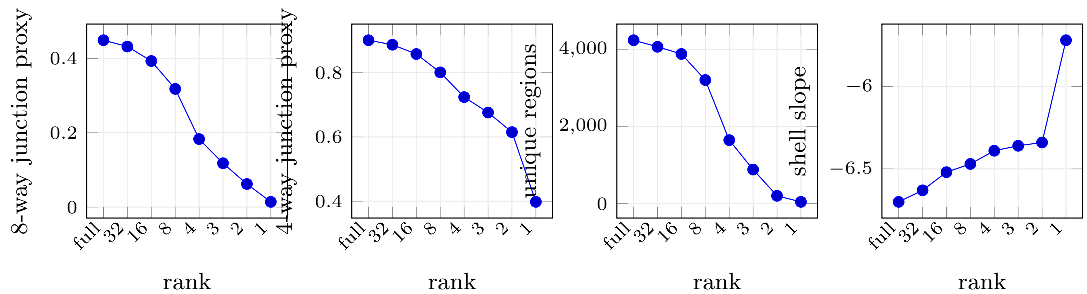

Figure 3: 3D CPWL validation. Rank constraints suppress high-codimension junctions before eliminating all regions, and the shell-averaged Fourier slope becomes visibly less steep.

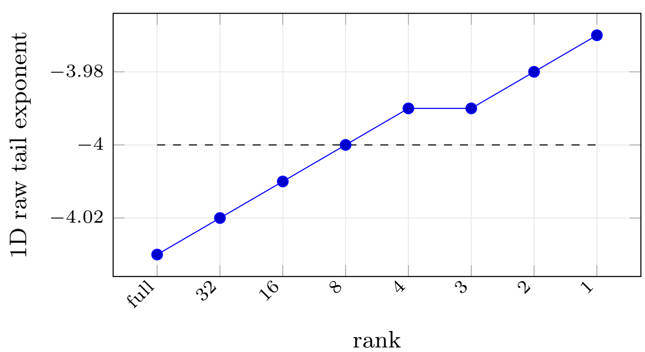

Figure 4: The 1D CPWL exponent remains pinned near −4. This negative result is theoretically expected and motivates measuring learned Fourier recovery instead.

Task	best r	best MSE	full MSE	best slope	full slope
Sparse	4	0.250	0.461	-0.29	-0.94
Dense	4	0.443	0.654	-0.29	-1.20

Table 1: The optimal rank is intermediate, not minimal. The same rank is selected by the approximation/optimization proxy in Fig. 5.

7.3 E3: width 2^{15} rank sweep in Fourier space

We train on two 1D Fourier targets: a sparse dyadic mixture $\{1, 2, 4, 8, 16, 32\}$ and a dense mixture with modes $1, \ldots, 32$. We sweep ranks 1 to 50 and compare against the full endpoint. The evaluation is deliberately Fourier-native: final MSE, recovery slope β_r, high/low recovery ratio, and the theoretical risk proxy (16).

7.4 E4: mode-wise recovery and training curves

The rank sweep should not only win in scalar MSE; it should visibly improve the Fourier scaling law. Figure 6 plots mode-wise recovery at the final training horizon. The full endpoint learns low modes and misses high modes, producing a steep negative recovery slope. Rank 4 maintains a much flatter recovery curve over the target bandwidth.

8 Connecting to Sobolev/Fourier training

The natural training loss for this theory is not only pointwise MSE. A Sobolev or Fourier-weighted loss

$$\mathcal{L}_s(f) = \sum_k (1 + |k|^2)^s |\widehat{f}(k) - \widehat{f}^\star(k)|^2 \qquad (20)$$

directly rewards high-frequency recovery. In the bound (16), this simply replaces $|\widehat{f}^\star(k)|^2$ by $(1 + |k|^2)^s |\widehat{f}^\star(k)|^2$. The recent Sobolev-training scaling results of Yang and He give a complementary framework for optimizing width/depth under Sobolev loss; here the new knob is rank, which changes the Fourier transfer and the effective dimension. The program is therefore to combine: approximation rates for shallow/deep ReLU networks, Sobolev generalization rates, and the rank-dependent spectral transfer $\lambda_r(k)$.

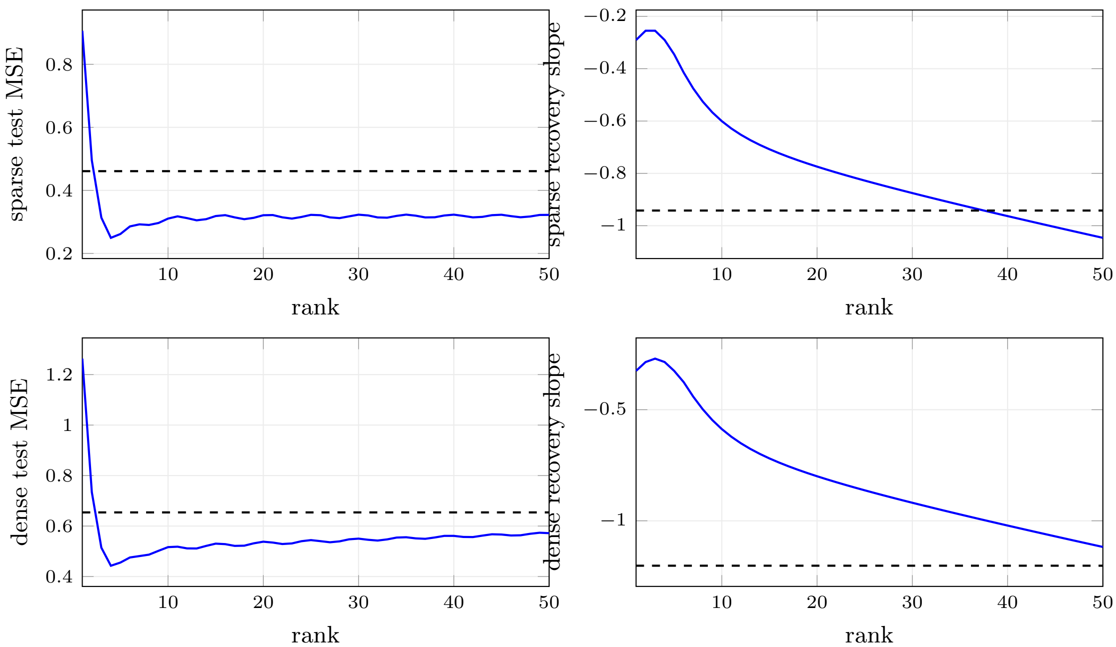

Figure 5: Width-2^{15} Fourier/kernel rank sweep. Solid curves sweep ranks $1, \ldots, 50$; dashed lines are the full-rank endpoint. The same intermediate rank lowers finite-budget MSE and makes the learned Fourier recovery scaling visibly flatter. High/low recovery and the bound decomposition are shown in the appendix.

9 Interpretation

The experiments support the refined conjecture.

- Low rank does not merely compress parameters. It changes the CPWL complex by constraining path coefficients and switching normals.
- In dimensions where high-codimension faces exist, this reduces high-codimension face moments and flattens finite-shell spectra.
- In one dimension, the raw CPWL exponent remains near -4; the visible spectral improvement is a flatter learned Fourier transfer function.
- The optimal rank is intermediate. Ranks 1–2 are too restrictive; full rank is too spectrally biased at finite training time; rank 4 balances approximation and optimization in our width-2^{15} Fourier sweep.

10 Limitations and next tests

The current width-2^{15} result uses a kernel/random-feature limit for the full endpoint rather than storing a dense hidden matrix. This is the right computational object for a very wide model, but the next version should also include direct finite-width SGD runs at the largest feasible widths. Second, the face moments M_q are estimated through grid proxies; exact enumeration of high-dimensional CPWL complexes remains hard. Third, the bound (16) is currently a predictive proxy. A theorem would require proving rank-dependent lower and upper bounds on $\lambda_r(k)$ for the relevant architecture.

11 Conclusion

The low-rank spectral-bias story survives the 1D exponent caveat once it is stated in the right variables. Low rank constrains path space, path constraints constrain CPWL faces, face moments control Fourier shells, and finite-budget Fourier training has an intermediate optimal rank. The clearest empirical signature is not a raw 1D exponent shift; it is the visibly flatter recovery scaling law and lower finite-budget error at the optimal rank.

Acknowledgment of scope

This main paper intentionally includes more derivations and plots than a four-page workshop version would keep. A submission version can compress the path-space derivation, move the full rank-sweep panels to the appendix, and retain the CPWL mechanism plus the Fourier recovery figure.

References

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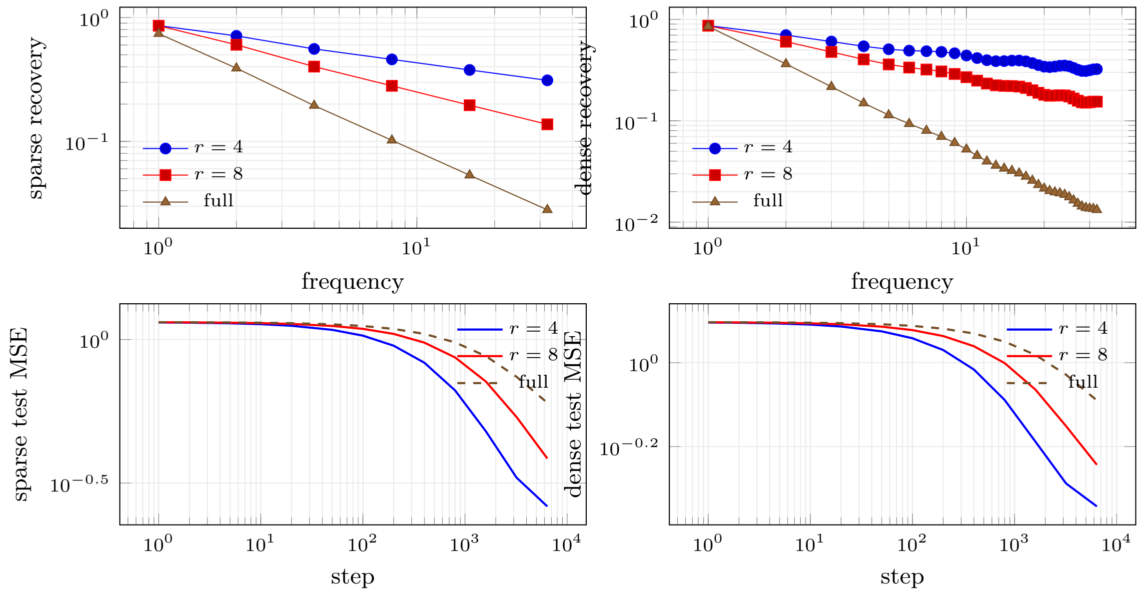

Figure 6: Fourier-space validation. Top: rank 4 gives a visibly flatter recovery scaling law than the full endpoint. Bottom: the same rank reaches a lower finite-budget test error.

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